

MALEK HAMMOUD

malekhammoud.com

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EDUCATION

- **Western University** London, ON
 - *Western's Initiative for Scholarly Excellence (WISE) – Concurrent Enrollment* Sep 2025 – Dec 2025
 - Selected for highly competitive dual-enrollment program for high-performing students.
 - **Grades:** Computer Science I (**97%**), Intro to Data Science (**92%**).
- **London Central Secondary School** London, ON
 - *Ontario Secondary School Diploma* Sep 2022 – Jun 2026
 - **Hack Club President:** Merged student clubs to form the Central Programming Alliance (50+ members).
 - Deployed a self-hosted DMOJ competitive programming platform on Azure to bypass school network blocks.

EXPERIENCE

- **Playtoon It, Inc.** Remote
 - *Full-Stack Developer Intern* Mar 2025 – Aug 2025
 - Collaborated with a 15-member engineering team to ship production features for a story platform.
 - Engineered optimized loading screens and integrated secure upload flows and profanity filtering.
- **Microsoft Sustainability Challenge** Via The Knowledge Society
 - *Technical Lead – Top 5 of 172 Teams* Mar 2025
 - Led technical research for a biomimicry-inspired cooling system saving 100M+ liters of water annually.
 - Presented technical feasibility and financial impact analysis directly to Microsoft executives.
- **Bryck Technologies** Remote
 - *Software Developer Intern* Sep 2024 – Jan 2025
 - Led the development of a full-stack application deployed to production for a 20+ person company.
 - Architected UI components using Storybook and managed PostgreSQL databases; contributed to daily Agile standups.

TECHNICAL PROJECTS

- **Flow Arch (Linux OS) — SourceForge Rising Star Award** flowarch-os.github.io
 - *Lead Developer & Maintainer* Dec 2025 – Present
 - Engineered a custom Arch Linux-based OS with **300+ downloads**; built using Arch ISO and Calamares installer.
 - Awarded the **SourceForge Rising Star Award** for rapid community adoption.
- **Basil Home AI (Startup) — Hack the 6ix Winner** basilhome.ca
 - *Founder & Lead Developer* Jul 2025 – Present
 - Building an AI-powered smart refrigerator startup; configured QNX RTOS on Raspberry Pi for embedded control.
 - Trained YOLOv5 models for real-time grocery detection and integrated Gemini API for recipe generation.
 - Secured victory at Hack the 6ix (Deloitte's Best Use of AI for Green) by demonstrating market viability.
- **GreenGuardian — Canada Wide Science Fair Bronze** View Project
 - *Sole Developer* Nov 2023 – May 2024
 - Designed an autonomous agricultural robot using computer vision to detect and target invasive weeds.
 - Developed custom pathfinding algorithms and 3D-printed chassis; successfully demoed at national level.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript/Node.js, Bash, SQL, HTML/CSS

Tools & Systems: Linux (Arch, Debian), QNX, Docker, React, Express, WASP, Storybook, PostgreSQL, Git, YOLOv5

HONORS & AWARDS

- **Hackathons & Industry**
 - **Hack the 6ix** – Winner, Deloitte's "Best Use of AI for Green" (2025)
 - **Hack the North** – Top 32 Finalist (2025)
 - **TKS Global Hackathon** – Top 4 Winner (2024)
- **Academic & Computing**
 - AP Computer Science A – Highest Score (5/5) (2025)
 - Canadian Computing Competition – Senior Distinction (2024)